

MDSA20030 - Endocrine Biology

Growth Hormone / Insulin Like Growth Factor – I Axis



Learning Objectives

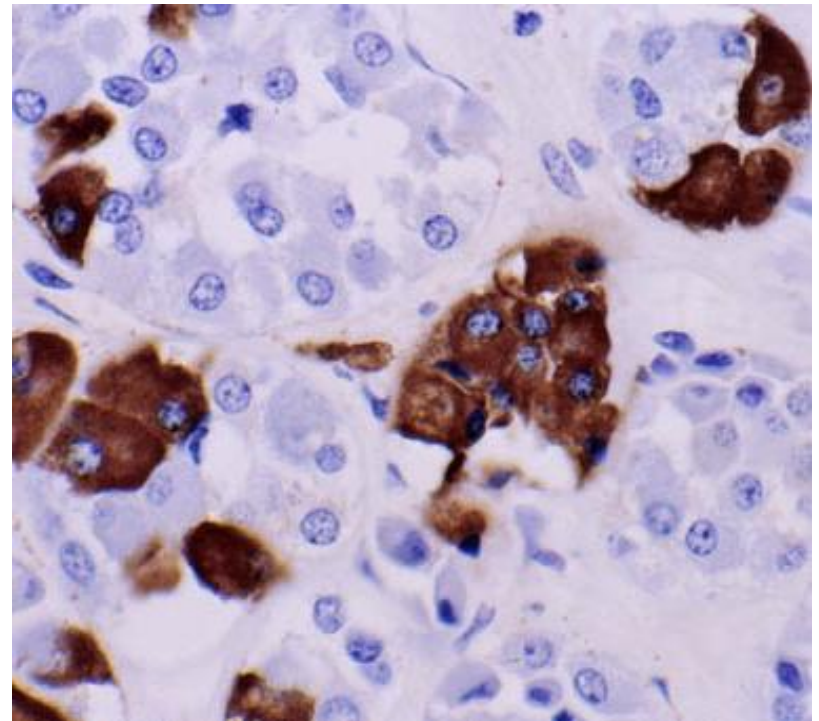
To understand

- factors influencing pituitary GH secretion
- the physiological actions of GH
- the GH/IGF-I axis
- clinical conditions of GH deficiency and excess ie. GH deficiency and acromegaly
- Possible role of GH in the aging process and the potential for GH therapy

Growth Hormone

- Evolutionary conserved peptide
- Anterior pituitary (somatotroph cells)
 - large prominent secretory granules
 - 5-15mg (4-8% of dry weight of pituitary)
 - extrapituitary synthesis
- Essential for linear growth in postnatal life (but plays no role in fetal growth)
- Important metabolic actions

Immunohistochemical staining of somatotrophs with anti GH antibody



Regulation of GH Secretion

- Changes in GHRH and SST secretion in hypothalamus

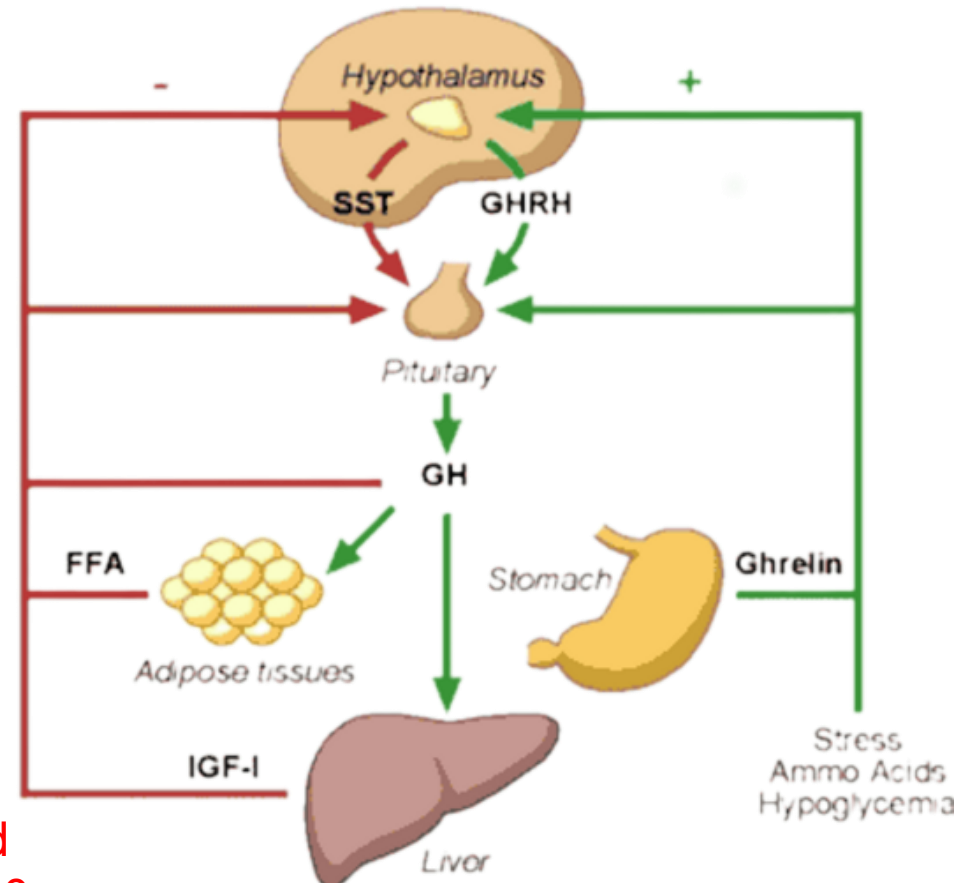
- IGF-I (long –ve feedback loop)
- GH itself (short –ve feedback loop)
- some AAs (eg. Arg) $\uparrow$ GH
- NEFAs $\downarrow$ GH
- hypoglycemia $\uparrow$ GH**
- ghrelin (stomach) $\uparrow$ GH

- Exercise

- vigorous exercise (depletion of energy reserves) $\uparrow$ GH

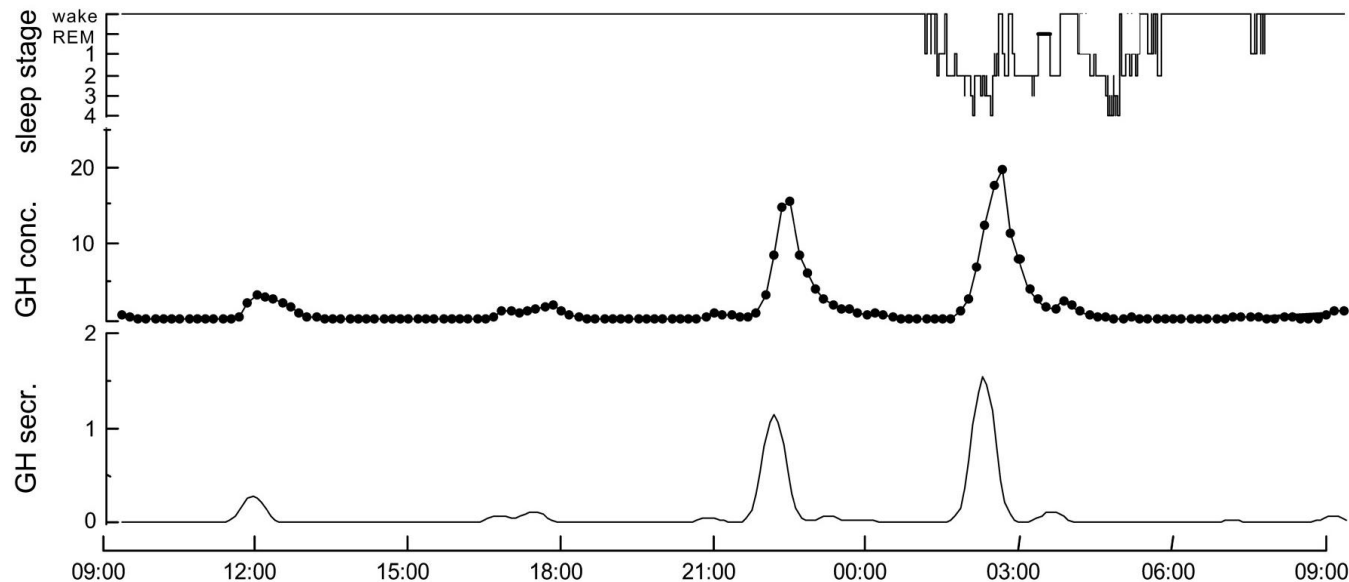
- Somatopause**

- normal age related decline in GH (and IGF-I) of ~ 50% between ages of 20-60**



Pulsatile and Circadian Pattern of GH Secretion

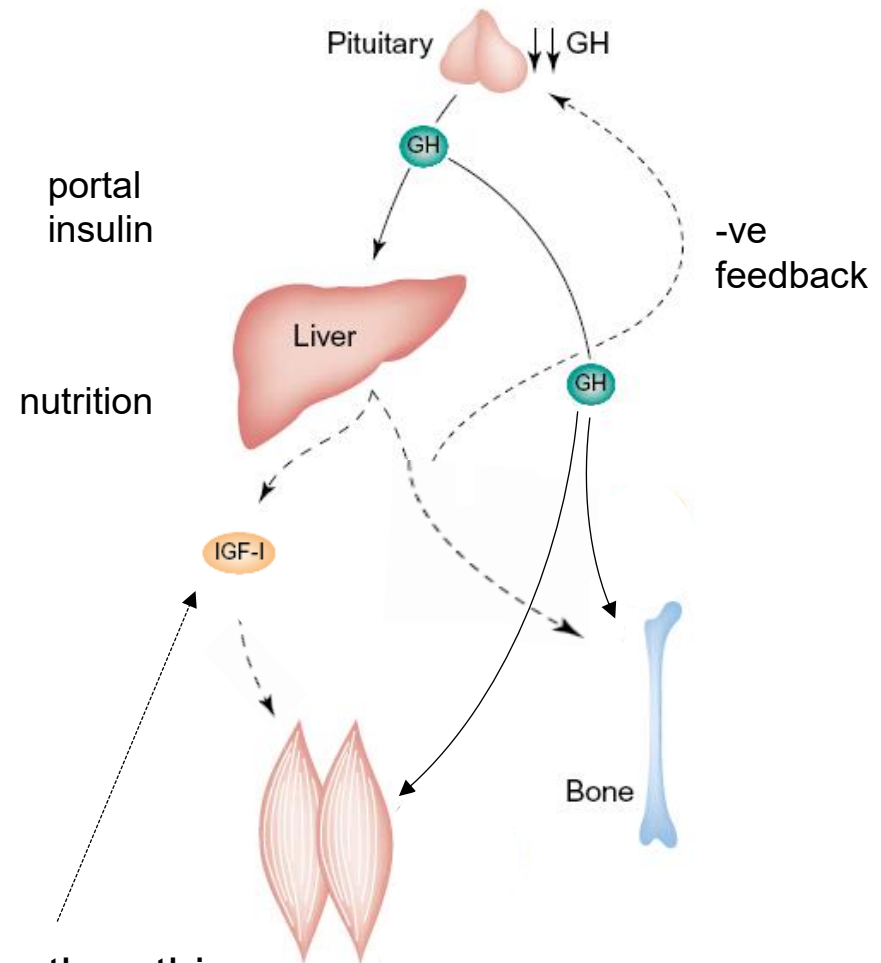
- Pulsatile secretion
 - regular pulses at 3-4h intervals
- Circadian rhythm
 - peak secretion coincides with onset of deep sleep (REM4)
 - 70% daily GH secretion



This pattern of regulation is essential for normal growth

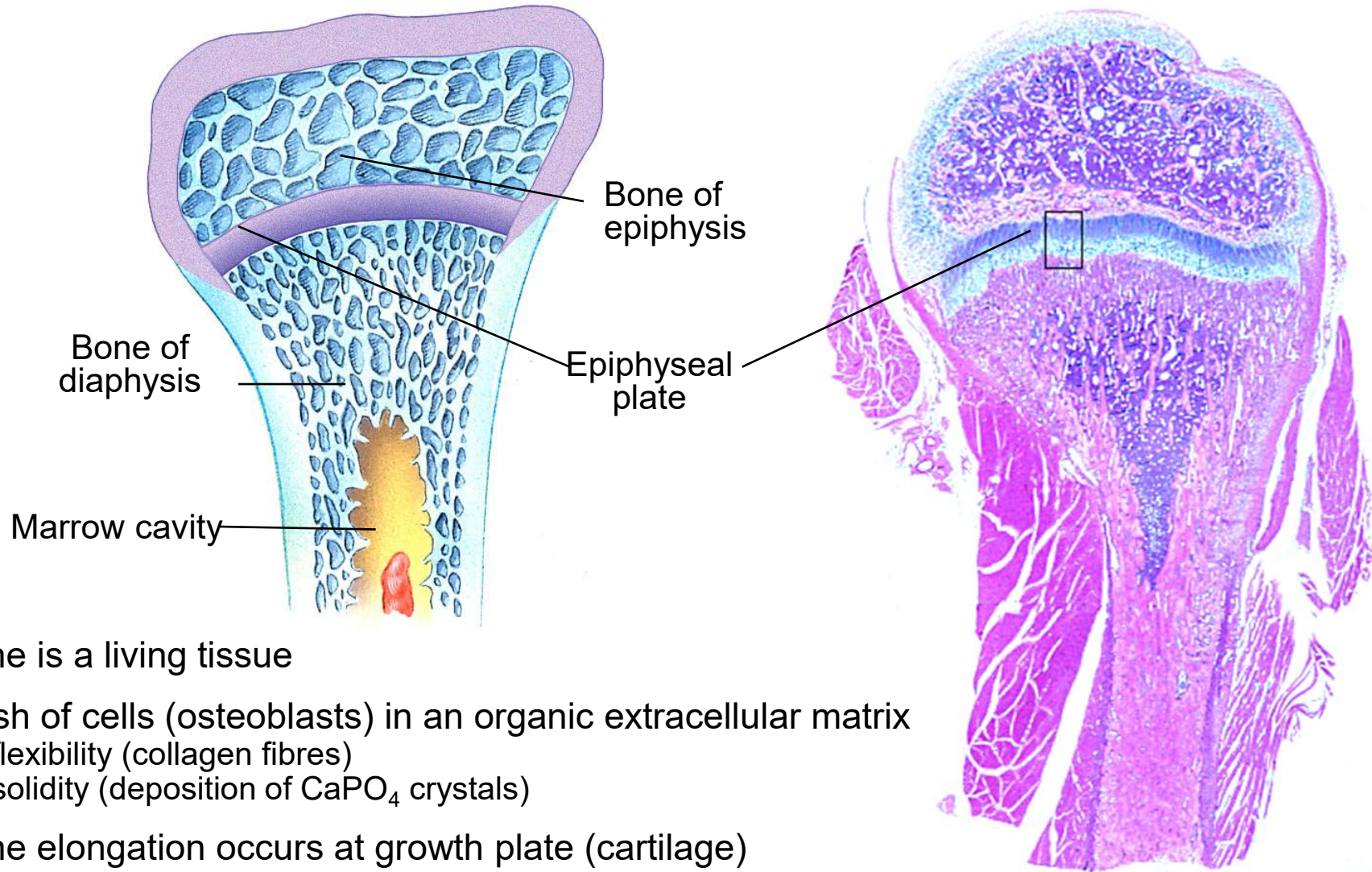
GH/IGF-I Axis

- Anabolic (growth promoting) effects of GH partly mediated by another hormone, IGF-I
 - produced in liver in response to GH
 - acts in a classical endocrine fashion
- Essential endocrine axis in growth
 - GH and IGF-I have independent and interdependent growth promoting actions ('dual effector hypothesis')
- and metabolism
 - GH is a contrainsulin hormone
 - IGF-I has insulin like metabolic actions

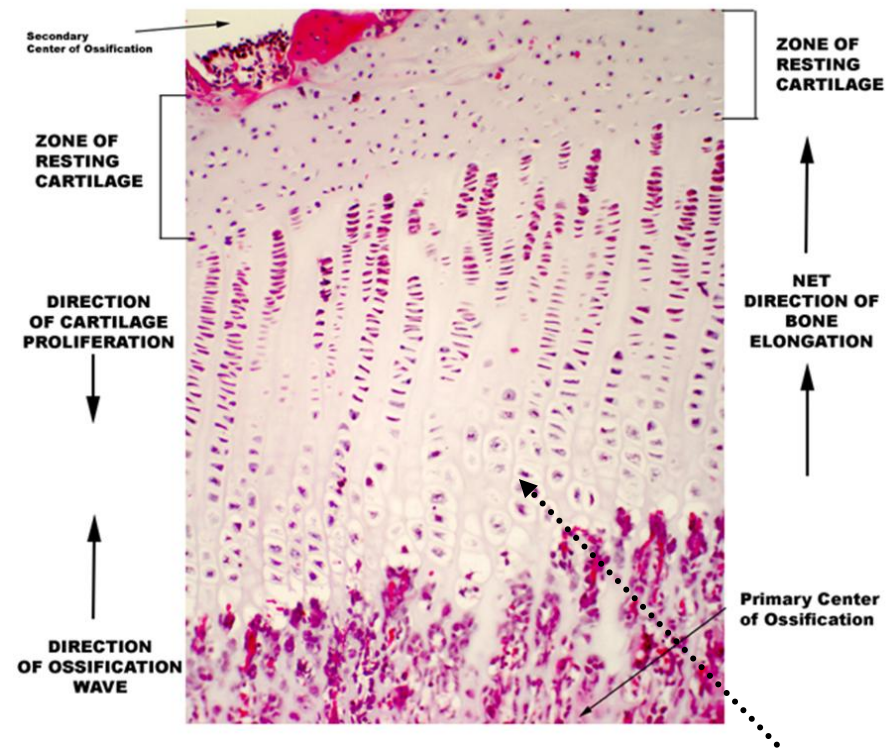
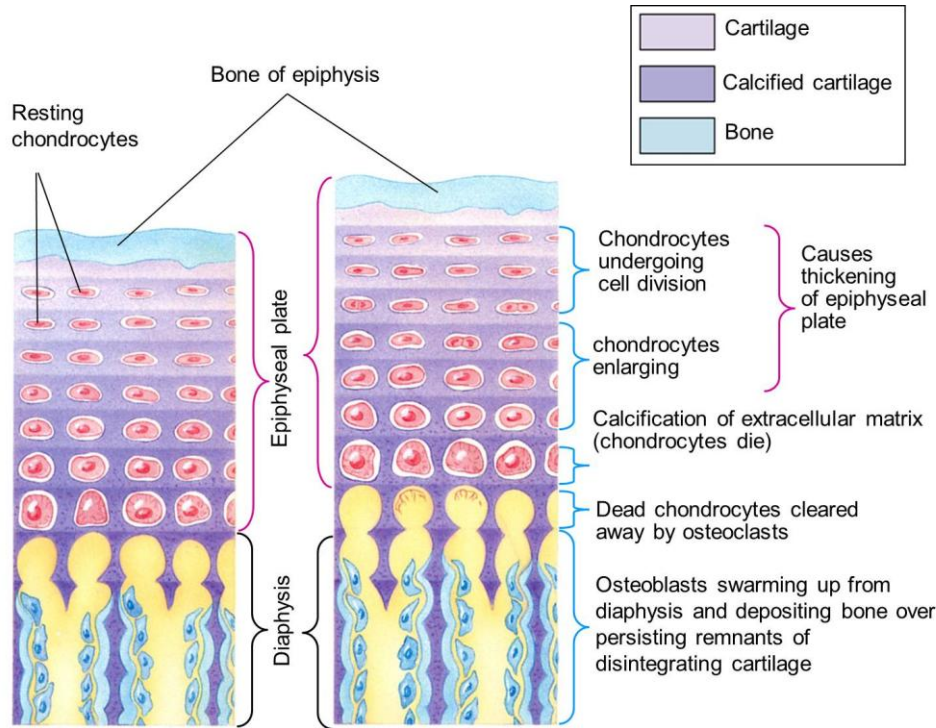


More complex than this,
more later

Linear Growth Requires Longitudinal Bone Growth

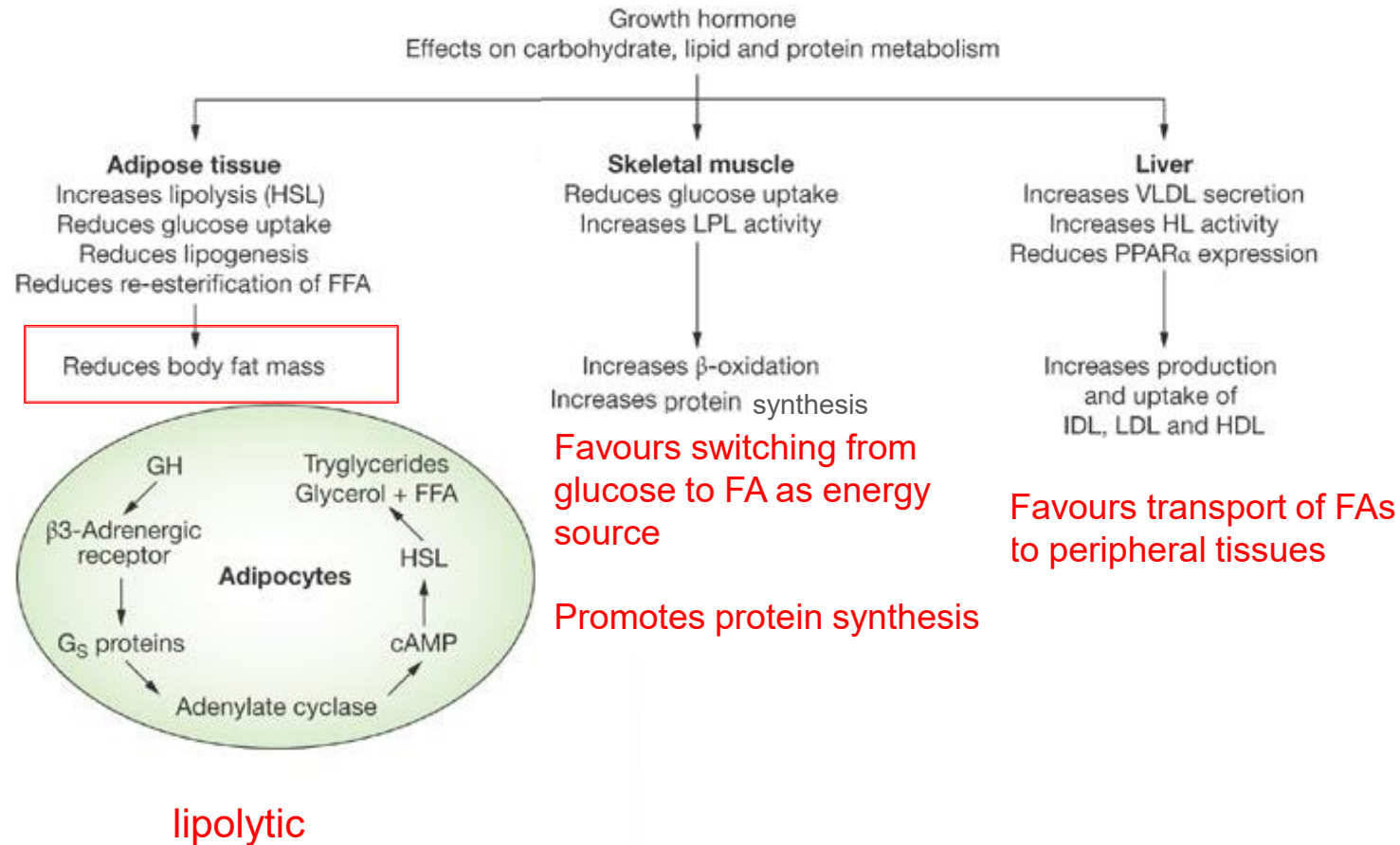


How Bone Grows



- Differentiation and proliferation of chondrocytes – controlled by GH and IGF-I respectively
- Mineralisation (closure of growth plate)
- Once growth plate closed, no further linear bone growth possible at this site

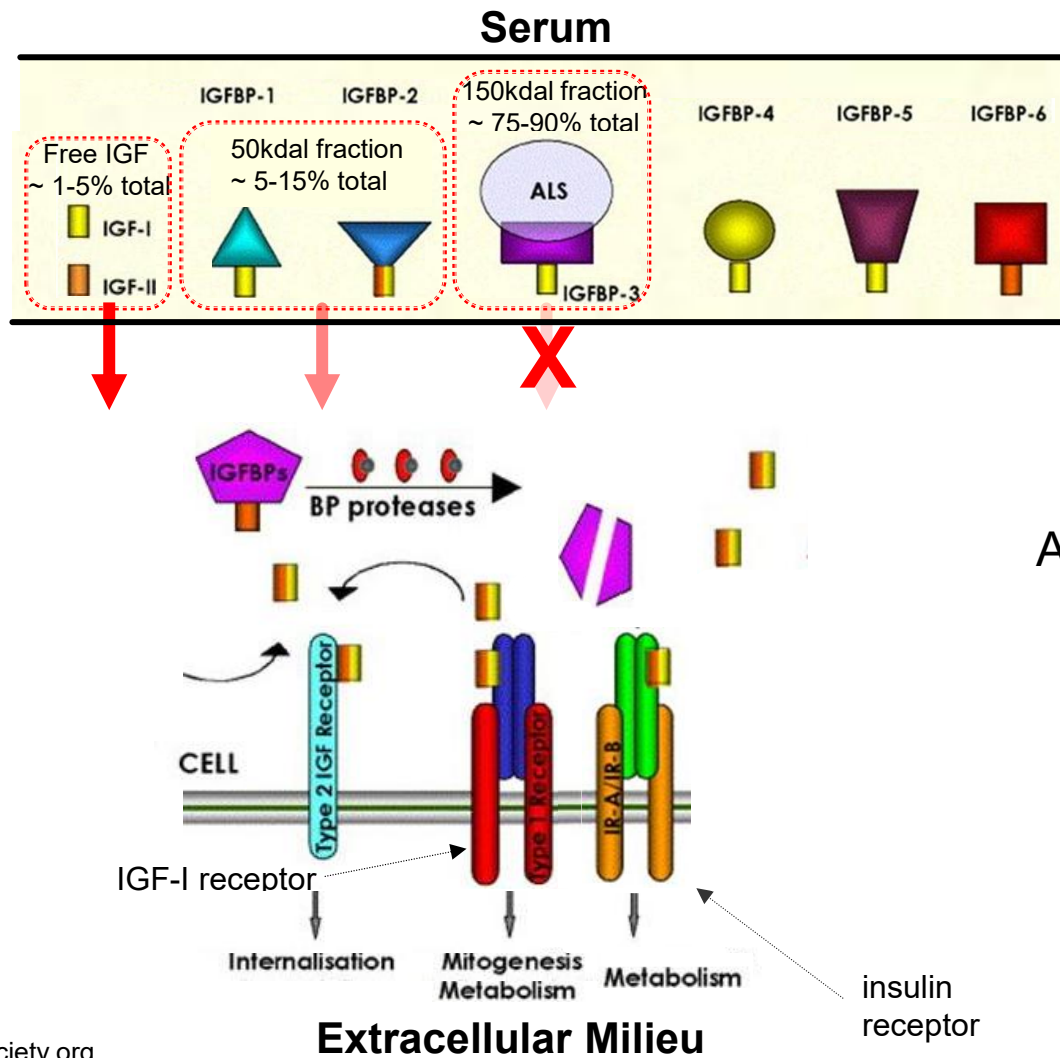
Metabolic Actions of GH



Insulin Like Growth Factor-I

- Circulating IGF-I derived mainly from liver (GH dependent) - endocrine
 - produced in many extrahepatic tissues in GH independent fashion - paracrine
- Regulation
 - pituitary growth hormone (GH)
 - nutritional status
 - insulin (↓ hepatic IGF synthesis)
 - somatopause - ↓ GH and IGF-I >50% between ages of 20 and 60
- Anabolic actions
 - GH dependent and independent actions to promote fetal and postnatal growth
- Metabolic actions
 - ↑ glucose uptake/utilisation in skeletal muscle
 - ↑ protein synthesis in skeletal muscle
 - ↑ insulin sensitivity
 - ↓ GH secretion (-ve feedback regulation)

IGF Ligands, Receptors and Binding Proteins (IGFBP)



A complex story

Other Hormones Important in Optimal Growth

- Thyroid hormone (permissive for GH)
 - delayed/impaired growth in hypothyroid children
 - hypersecretion does not cause excessive growth
- Insulin (stimulates protein synthesis)
 - deficiency often impairs growth (type 1 diabetes)
 - hyperinsulinism often spurs excessive growth (eg. macrosomia in fetal life)
- Androgens
 - play role in pubertal growth spurt, anabolic actions in many organs
 - effects depend on presence of GH
- Estrogens
 - effects of estrogen on growth prior to bone maturation are not well understood

Little and Large, the Extremes of Growth

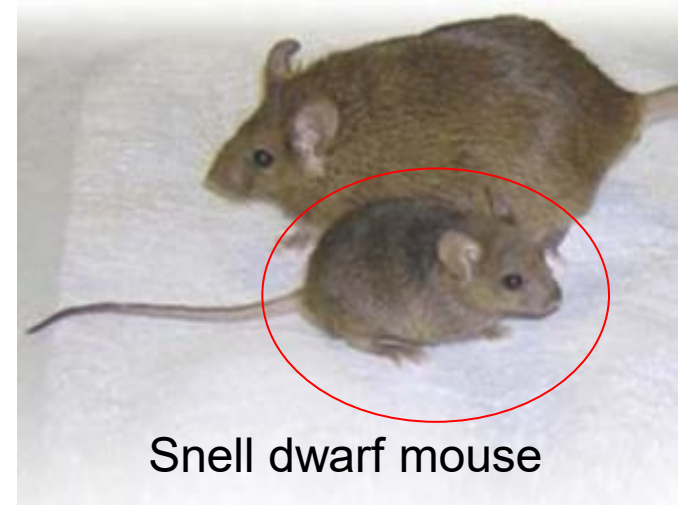
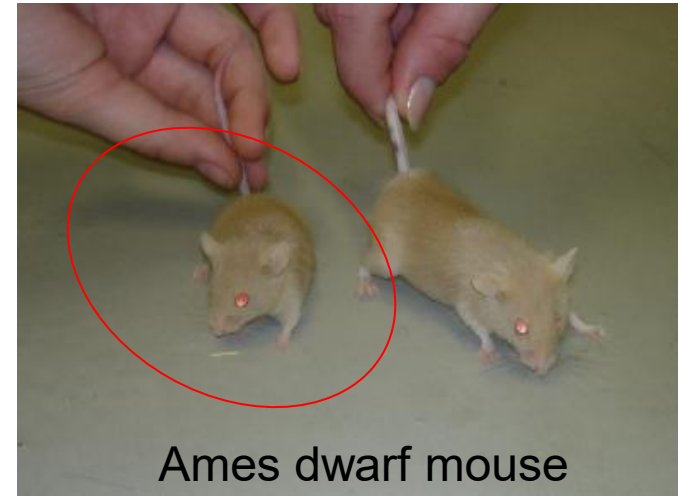


Leonid Stadnyk (8' 4")



GH Deficiency in Hypopituitary Dwarf Mice

- Spontaneous occurring dwarf mouse strains
- Abnormal pituitary cell differentiation due to mutations in
 - PROP1 (snell dwarf mouse)
 - Pit1(ames dwarf mouse)
- ↓ GH ↓ IGF-1
 - ↓ TSH and prolactin
- Sub or infertile
- Increased longevity (40-60%)
- Other models with increased lifespan
 - 'Little mice' (GHRH receptor mutation)
 - GHR deficient (Laron) mice

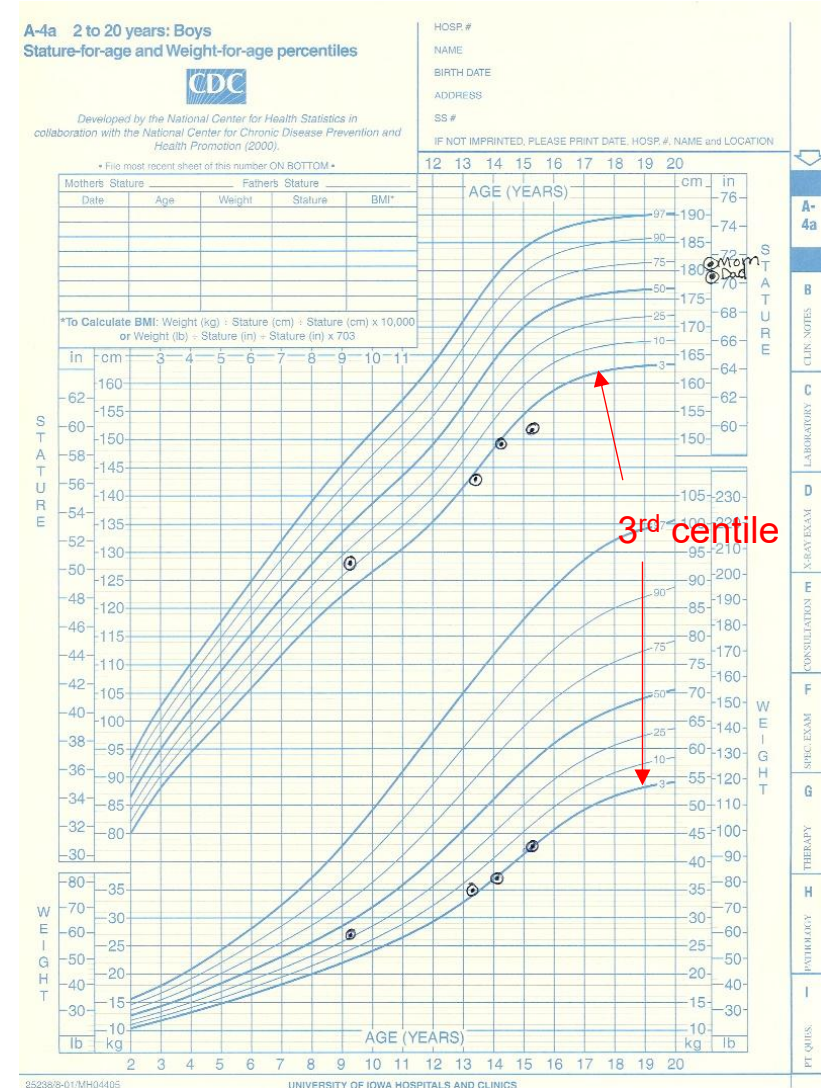


Congenital GH Deficiency

- Generally linked to hypopituitarism
 - birth trauma (eg. hypoxia)
 - midline defect syndromes (septo-optic dysplasia)
 - other congenital pituitary defects
 - empty sellar syndrome
 - pituitary stalk interruption syndrome (PSIS)
- GH deficiency may be isolated or may have panhypopituitarism
- Growth failure evident in early childhood but primary manifestations often metabolic
- ↓ GH ↓ IGF-1
- Treatment - GH replacement therapy

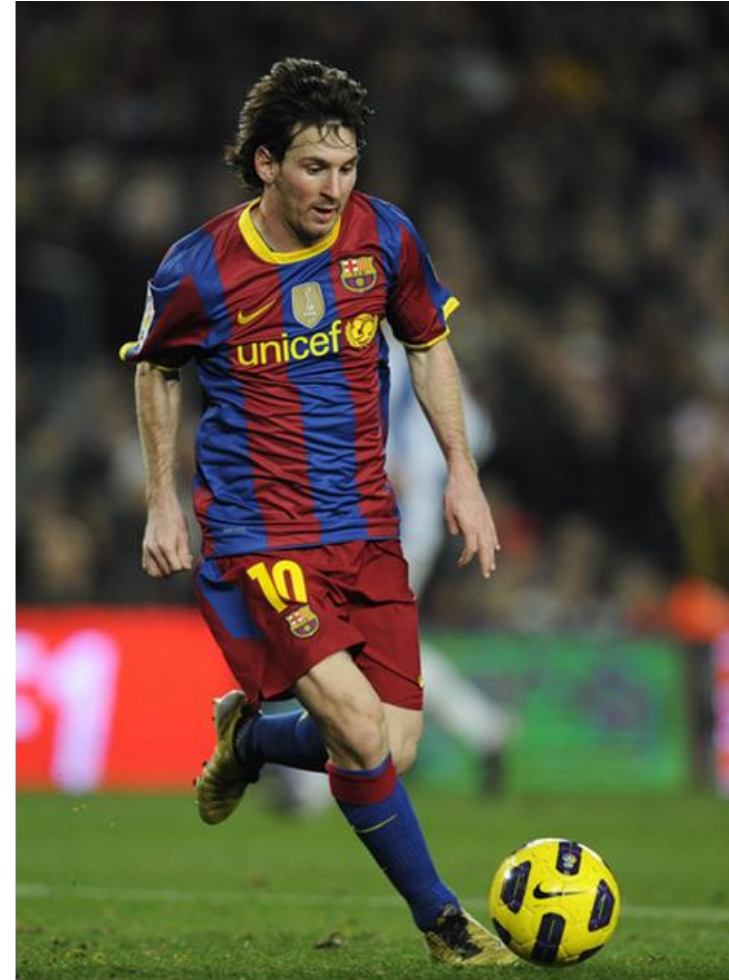
Growth Hormone Deficiency in Children

- Acquired growth hormone deficiency in childhood may result from
 - trauma
 - infections (eg, encephalitis, meningitis),
 - cranial irradiation
 - systemic diseases (eg. leukemia)
- Symptoms of GH deficiency in children include the following:
 - short stature
 - low growth velocity for age/pubertal stage
 - increased amount of fat around the waist
 - delayed tooth development
 - delayed onset of puberty



GH Replacement Therapy for GH Deficiency in Children

- diagnosed with GHD aged 11
- GH replacement therapy was costly and neither his family or club could afford it
- Barcelona offered to fund his treatment if he signed for them and moved to Barcelona
- the rest is history....
- Lionel Messi now measures 5' 7"



GH Resistance - Laron Dwarfism

COMMENTARY

Growth Hormone Receptor Deficiency in Ecuador*

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FIG. 1. Sisters 8.2 (left) and 6.1 (right) years of age with the appearance of GH deficiency, but with markedly elevated random GH levels. SDS for height were -8.3 and -9.4 with height ages of 15 and 11 months. A 10.5-yr-old sister (homozygous normal for the E180 splice mutation) had a height SDS of -0.3 as did the (proven) heterozygous father. Their mother's height SDS was -1.9 .

First described in 1966 by the Israeli paediatric endocrinologist Zvi Laron in Loja province of southern Ecuador

Observed significant growth failure in many children despite unusually high circulating GH (and low IGF-I) concentrations

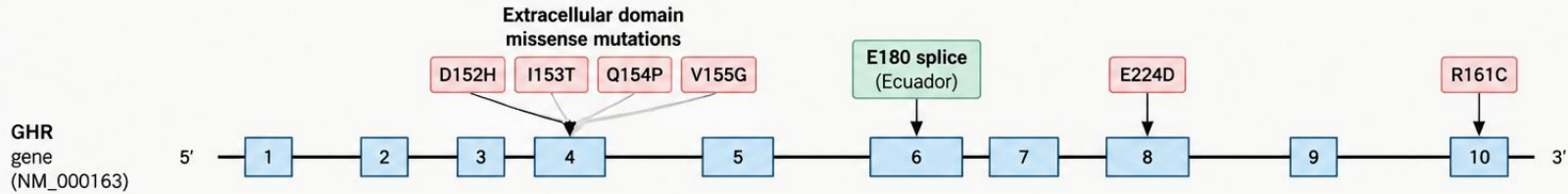
As well as short stature, exhibited poor muscle development, obesity and hypoglycemia

The condition was found to be a recessive disorder due to a mutation in the GHR gene

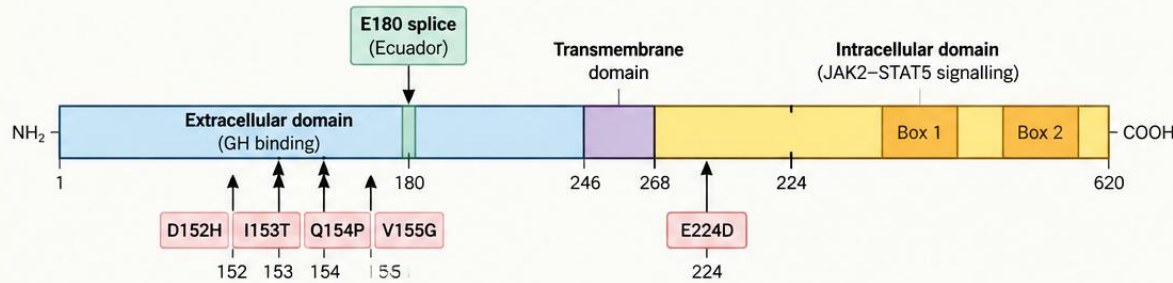
May be treated with IGF-I

Mutations in the Growth Hormone Receptor (GHR) in Laron Syndrome

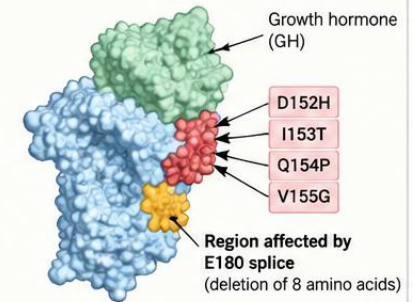
A. GHR gene (chromosome 5p13) and common mutations in Laron syndrome



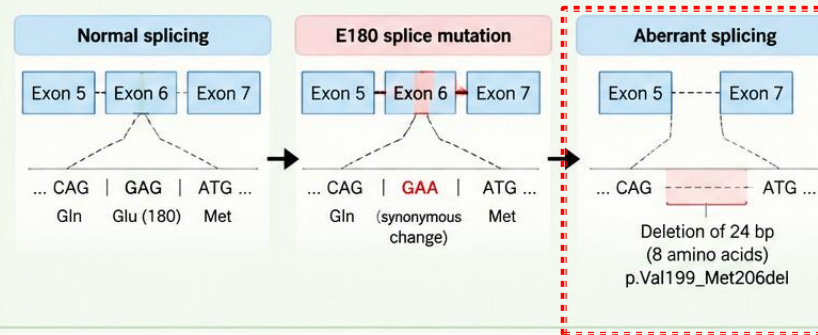
B. GHR protein domains and location of mutations



C. Structure of the extracellular domain



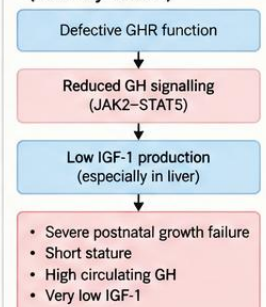
D. The Ecuadorian E180 splice mutation



E. Functional consequences of GHR mutations

Mutation (example)	Type of change	Effect on receptor
D152H I153T Q154P V155G	Missense (substitution of single amino acid)	- Impaired GH binding - Reduced cell-surface expression/trafficking - Abnormal dimerisation - Reduced signalling (JAK2-STAT5)
E180 splice (Ecuador)	Synonymous nucleotide change creating a cryptic splice site	- Deletion of 8 amino acids from extracellular domain - Impaired GH binding - Markedly reduced receptor function
E224D R161C (others)	Missense	Variable effects on expression, ligand binding and signalling

F. Resulting phenotype (Laron syndrome)



Researchers have mapped the E180 mutation back to a single ancestor from the Iberian Peninsula. Sephardic Jewish individuals fleeing the Spanish Inquisition migrated to the New World, carrying the E180 splice variant into South America, where it became heavily concentrated through geographic isolation.

Acquired GH Resistance

Condition	Principal mechanism	Typical endocrine findings
Malnutrition or prolonged fasting	Reduced hepatic responsiveness to GH, conserving energy and limiting growth	Increased GH, reduced IGF-1
Chronic inflammatory disease	Cytokines, particularly IL-6 and TNF- α , interfere with GH–JAK2–STAT5 signalling	Reduced IGF-1 despite preserved or increased GH
Chronic liver disease	Impaired hepatic IGF-1 production and sometimes reduced GH responsiveness	Low IGF-1, often elevated GH
Poorly controlled type 1 diabetes	Portal insulin deficiency reduces hepatic GH receptor expression and signalling	Elevated GH, low or inappropriately normal IGF-1
Severe systemic illness	Inflammatory and metabolic adaptations alter GH secretion and peripheral responsiveness	Variable GH, generally reduced IGF-1

Adult Onset GH Deficiency (AGHD)

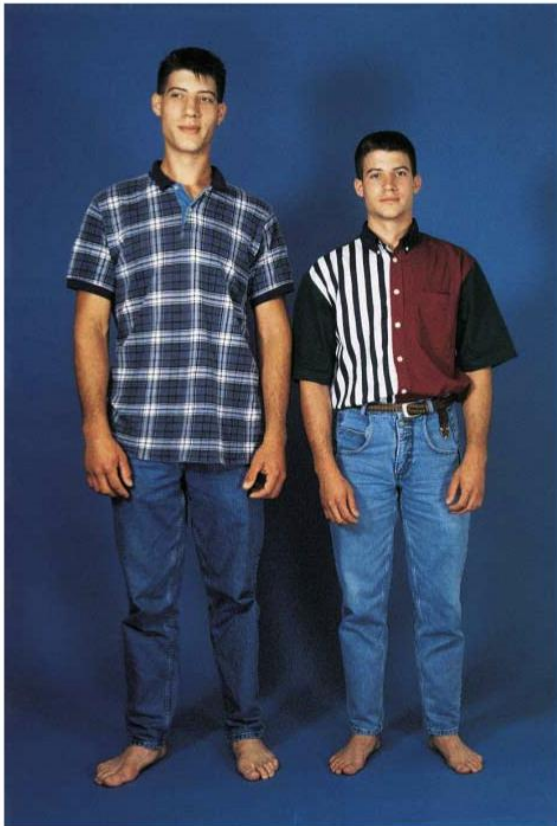
- Main causes
 - pituitary tumours that impair somatotroph function
 - head trauma
 - vascular abnormalities of hypophyseal circulation
- Main features
 - Remember GH is important in maintenance of skeletal muscle and bone and in glucose homeostasis
 - metabolic abnormalities
 - altered body composition (↓ lean body mass, ↑ adiposity)
 - Osteoporosis
- GH replacement therapy effective

GH Hypersecretion

- Congenital (or pre-puberty)
 - acromegalic gigantism (GH excess prior to fusion of epiphyseal growth plate)
- Acquired (adult onset)
 - bone thickening (not lengthening) but rare cases of gigantism reported
 - metabolic abnormalities similar to type 2 diabetes
 - increased risk cardiovascular disease, cancer
 - often simply referred to as acromegaly
- Main causes
 - functioning pituitary adenoma of somatotroph origin

Acromegalic Gigantism

Identical twins, 22 years old, excess GH secretion



A

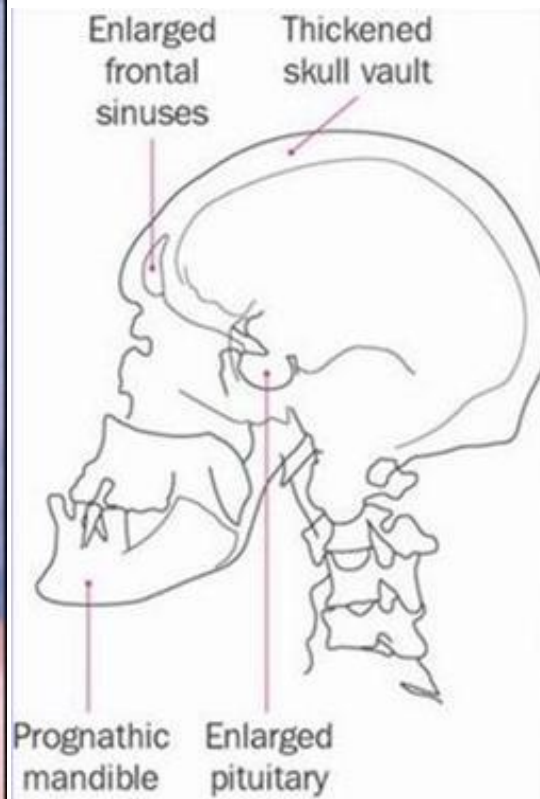


B



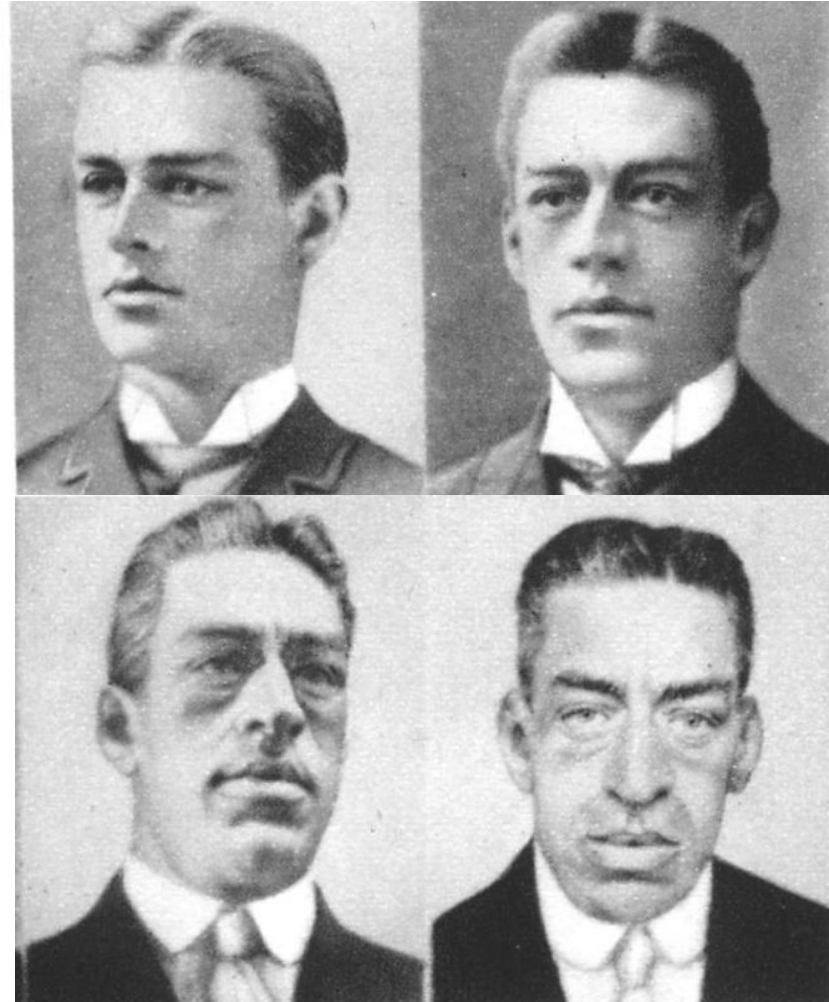
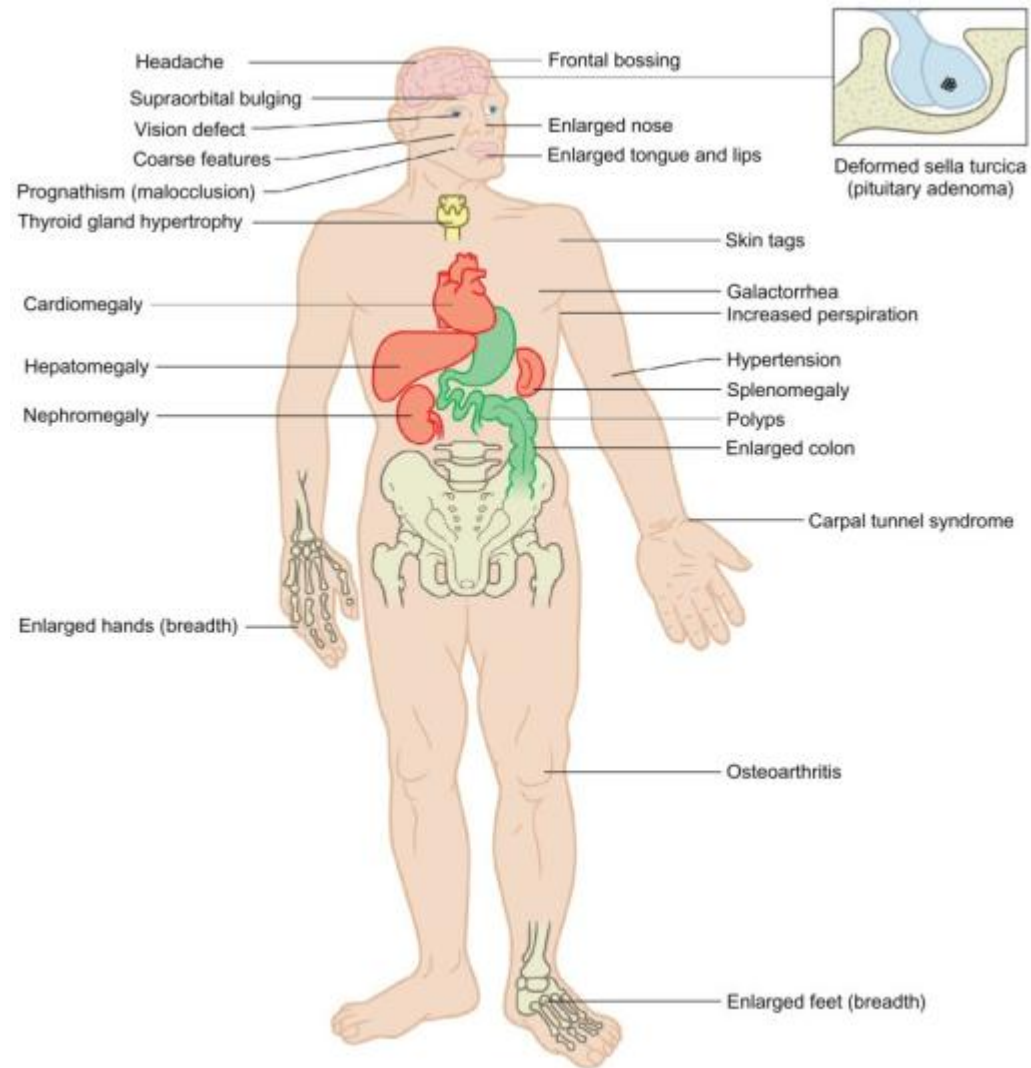
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Teeth gapping

Acromegaly: Adult Onset GH Hypersecretion



Adult Onset Acromegalic Gigantism

<http://www.channel4.com/programmes/bodyshock/4od#3105551>

died aged 34 in Jan 2013

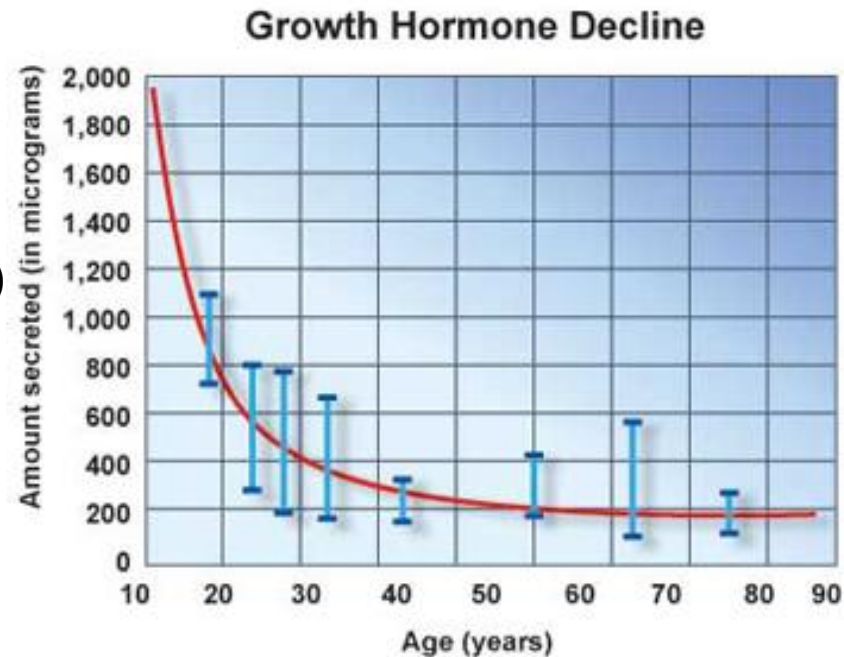


10 years

7ft 2 in in height
weighed 400lbs

Age-related Changes in Body Composition and Function

- Normal Aging
 - loss of lean body mass (sarcopenia), decreased strength, loss of functional capacity
 - increase in total fat mass (% body fat)
 - insulin resistance (type 2 diabetes)
 - dyslipidemia ($\uparrow$ LDL cholesterol, TGs, FAs)
- Adult GH/IGF-I deficiency
 - central adiposity
 - decreased muscle mass
 - decreased bone density
 - decreased insulin sensitivity and increased susceptibility to diabetes



notice the similarity ??

**New! Doctor
Recommended**

The Reverse Aging Miracle

RELEASE YOUR OWN GROWTH HORMONE AND ENJOY:

- Improved sleep & emotional stability
- Increased energy & exercise endurance
- Loss of body fat
- Increased bone density
- Improved memory & mental alertness
- Increased sexual potency
- Increased muscle strength & size
- Reverse baldness & color restored
- **Regenerates Immune System**

- Strengthened heart muscle
- Controlled cholesterol
- **Normalizes blood pressure**
- Controlled mood swings
- Wrinkle disappearance
- Reverse many degenerative disease symptoms
- Heightened five senses awareness
- Increased skin thickness & texture

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Formula**

This program will make a radical difference in your health, appearance and outlook. In fact we are so confident of the difference GHR can make in your life we offer a 100% refund on unopened containers.

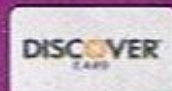
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Next

Tutorial 1 – Pituitary Physiology
Friday Sept 25th @ 10:00 in B004

Thyroid and Parathyroid Anatomy
Friday Sept 25th @ 12:00 in C005

Biochemistry Practical (GTT)
Friday Sept 25th @ 15:00 in E2.53/2.54